

Aviral Asthana

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OBJECTIVE

Machine Learning Engineer with hands-on experience building, optimizing, and evaluating end-to-end machine learning systems. Strong foundation in deep learning, computer vision, and applied machine learning with proficiency in Python and SQL. Experienced in working across the full ML lifecycle including data preparation, feature engineering, model development, evaluation, and performance optimization. Passionate about applying ML techniques to solve real-world problems with measurable impact.

TECHNICAL SKILLS

- **Machine Learning & AI:** Machine Learning, Supervised & Unsupervised Learning, Deep Learning, Convolutional Neural Networks (CNNs), Feature Engineering, Model Evaluation & Validation, Hyperparameter Tuning, Model Optimization, Recommendation Systems
- **Computer Vision:** YOLOv7, OpenCV, MediaPipe, Image Classification, Object Detection, Video Analytics, Real-time Inference
- **Programming & Tools:** Python, SQL, Data Structures & Algorithms, Object-Oriented Programming (OOP), Git, Linux, Jupyter Notebook
- **Cloud, MLOps & Systems:** Cloud Computing (AWS – EC2, S3, IAM), Model Deployment, Scalable ML Pipelines, REST APIs (Foundational), Model Monitoring
- **Data & Analytics:** Relational Databases, ETL Pipelines, Data Modeling, Query Optimization, Performance Tuning
- **Tools:** Git, Linux, Jupyter Notebook

PROFESSIONAL EXPERIENCE

Trinetradhar Info Solution Pvt. Ltd. | *Project Trainee.*

Nov 2023 - Jan 2024

- Designed, benchmark-tested, and optimized a relational database for a government e-Reporting System, achieving ~25% faster query performance and ~30–40% lower data redundancy through schema normalization, indexing, and performance tuning under load.

Medium | *Author*

Aug 2022 - Present

- I write about science and cosmology to make complex ideas accessible. My most-read article reached ~180 views with a ~47% read ratio, reflecting strong engagement driven by clear explanations and visual reasoning.

PROJECTS

DAWG – Stray Dog Detection System

- Developed a YOLOv7-based real-time object detection system achieving ~85–90% detection accuracy at 25–30 FPS.
- Planned the solution for practical deployment, focusing on robustness and low-latency inference.

AI-Based Career Counsellor

- Built a machine learning-based recommendation system to suggest personalized career paths.
- Improved recommendation relevance by ~35% through feature engineering and model tuning.

Alzheimer's Disease Detection using CNNs

- Implemented a CNN-based medical imaging classification model achieving ~90% accuracy.
- Lowered false negatives by ~20%, supporting more reliable clinical decision-making.

FRUITIPEDIA – Automated Fruit Grading System

- Engineered a CNN-based classification system achieving ~88–92% accuracy across multiple fruit categories.
- Reduced manual inspection effort by approximately 50%.

American Sign Language (ASL) Gesture Recognition System

- Built a real-time ASL gesture recognition pipeline with <100 ms inference latency.
- Achieved ~85% accuracy using CNNs, OpenCV, and MediaPipe.

EDUCATION

Vellore Institute of Technology

Aug 2021 - Jul 2025

Bachelor of Technology (B.Tech.), Computer Science with specialization in Artificial Intelligence and Machine Learning (GPA: 8.64/10)

- **Achievements:** Participated in Omdena local project on monitoring water quality in Bhopal using satellite imagery and GIS techniques
- **Coursework:** Deep Learning, Natural Language Processing, Computer Vision, Artificial Neural Networks

CERTIFICATIONS

- AWS Certified Cloud Practitioner — Amazon Web Services (2024)
- Database Developer on Oracle Cloud — Oracle (2024)
- Applied Machine Learning in Python — University of Michigan (2023)
- Complete Python Bootcamp — Udemy (2022)

EXTRACURRICULAR & LEADERSHIP

- Core Member, Content Team — Hindi Club, VIT
- Core Member, Dev Code Community (Content & Events)
- Organizer & Host: Dev Talks Podcast and CodeHack Hackathon
- Attendee, IIT Delhi Rendezvous — Interacted with industry professionals